

Lab 29 — NumPy Statistics and Aggregation

Aim

To compute descriptive statistics such as mean, sum, standard deviation, minimum, and maximum using NumPy aggregation functions.

Algorithm / Procedure

1. Import the NumPy library.
2. Create a 2D NumPy array.
3. Calculate the overall mean.
4. Calculate the sum along columns and rows.
5. Calculate the standard deviation.
6. Find the minimum value of each row.
7. Find the overall maximum value.
8. Display all results.

Program

```
import numpy as np

a=np.array([[1,2,3],[4,5,6]])

print("Mean:",a.mean())

print("Sum along axis 0 (columns):",a.sum(axis=0))

print("Sum along axis 1 (rows):",a.sum(axis=1))

print("Std Dev:",a.std())

print("Min per row:",a.min(axis=1))

print("Max overall:",a.max())
```

Output

Mean: 3.5

Sum along axis 0 (columns): [5 7 9]

Sum along axis 1 (rows): [6 15]

Std Dev: 1.707825127659933

Min per row: [1 4]

Max overall: 6

Result

The NumPy statistical and aggregation functions were successfully implemented to calculate descriptive statistics along different axes of the array.